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Unit 3 - Programming the Butterfly Display

PDF Documents

PDF Document 1: 1738215716305-Chapter 8 - Unit 3 - Programming the Butterfly Display.pdf

This PDF document is included in the unit materials.

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UNIT 3 PROGRAMMING THE BUTTERFLY DISPLAY

Project Introduction

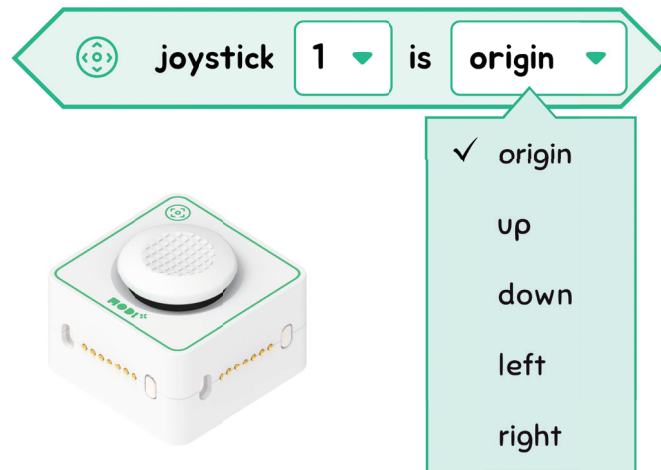
In this project, you are going to build a car that moves based on the direction you control using a joystick. The car will use two wheels, two motors, and the MODI joystick to steer. You will connect the MODI modules together to create a system that allows the car to move forward, backward, left, and right, depending on which way the joystick is pressed. If the joystick is not pressed, the car will stop and stay still.

HOW THE CAR WILL WORK:

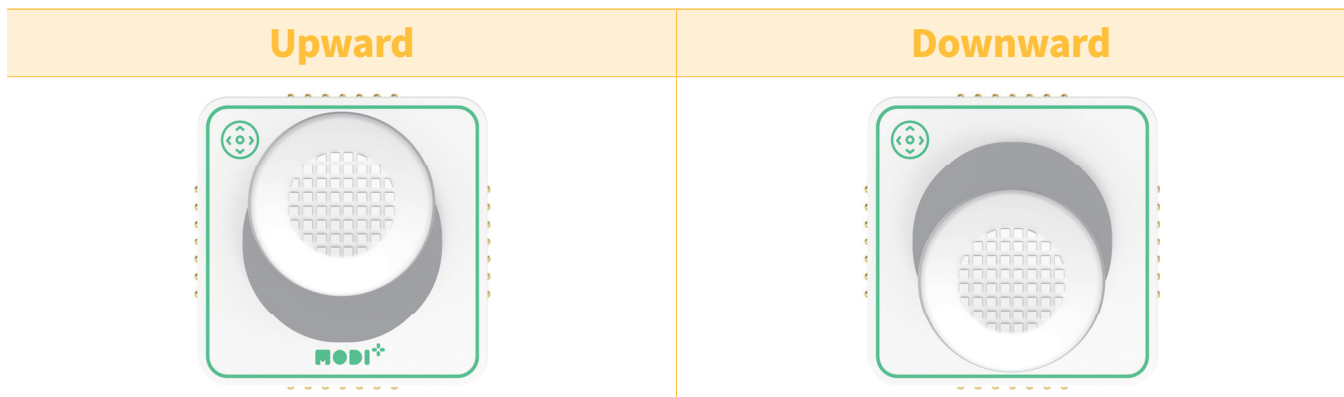
- JOYSTICK DIRECTIONS:** The joystick will have four directions—up, down, left, and right.
Up: Moves the car forward.
Down: Moves the car backward.
Left: Turns the car left.
Right: Turns the car right.
- MOTORS** The car will have two wheels powered by Motor A and Motor B. These motors will work together to make the car move in different directions.
Motor A controls the right wheel.
Motor B controls the left wheel.
- NO MOVEMENT WHEN IDLE** If the joystick is not being moved, the car will stay still. The car only moves when the joystick is pressed in one of the four directions.

Module Functionality

Let's look into the details of the Joystick module.

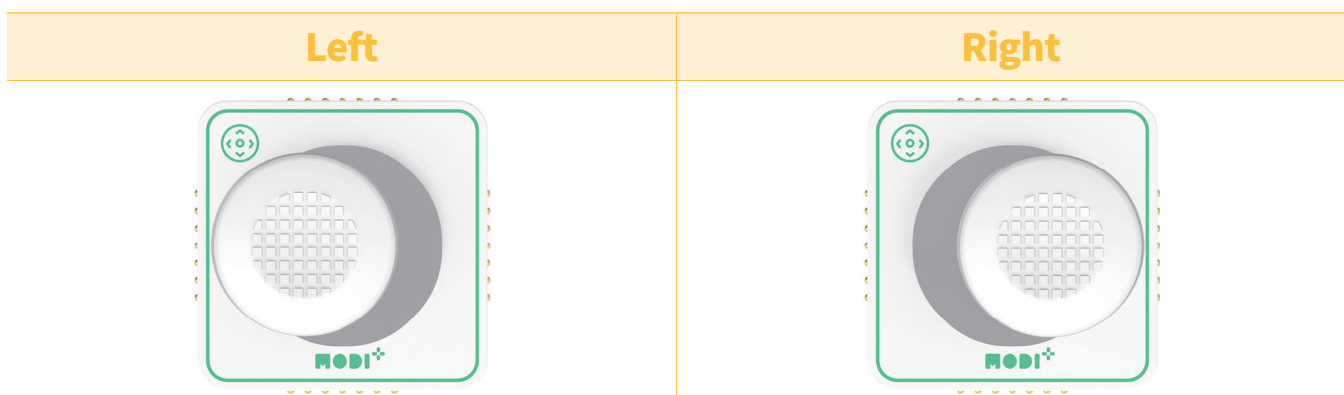


Including the neutral position 'origin,' the joystick has 5 different positions.



Pushing it upward and downward affects movement on the **y-axis**

- Range : -100 % (fully downward) ~ +100 % (fully upward)



Moving the joystick left and right controls movement along the **x-axis**

- Range : -100 % (fully left) ~ +100 % (fully right)



Activity: Fly Fly Butterfly

1 Materials Needed:



Network



Joystick



Motor A



Motor B

2 Code story

Motor A,B are stopped



When Joystick is up



Move forward

When Joystick is down



Move backward



When Joystick is right



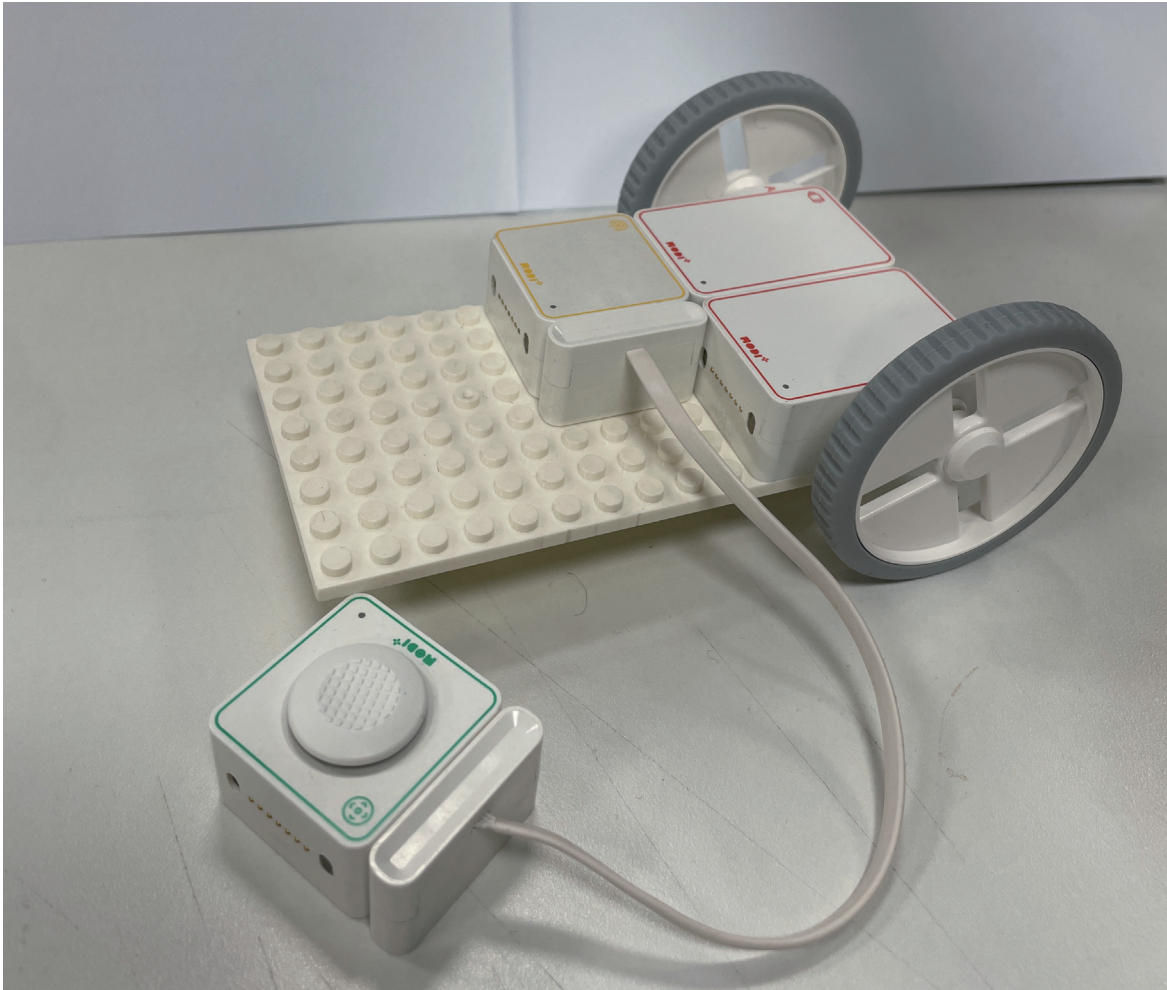
Move right

When Joystick is left

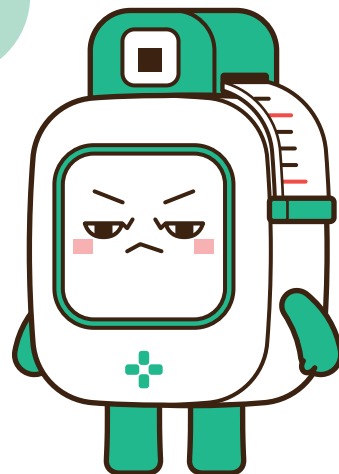


Move left

How does creation look like?



Use a module connector to connect Joystick and Network!





Code Quest



Checkpoint

What code is needed to make the car move left?

Hint: It should move the opposite way of turning right!